Automated Diabetic Retinopathy Prediction Using Feature-Based Machine Learning

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***Abstract*—Diabetic retinopathy is a leading cause of preventable blindness worldwide. This paper presents a feature-based machine learning approach to automated diabetic retinopathy prediction from retinal fundus images. We extract Gray-Level Co-occurrence Matrix (GLCM) texture features and use a Genetic Algorithm (GA) to simultaneously optimize LightGBM classifier hyperparameters and select the most relevant features. Our baseline model achieves 91.00% accuracy using all 18 GLCM features, while the GA-optimized model achieves 89.50% accuracy with only 4 features, a 77.8% reduction in feature set size. We evaluate the accuracy-complexity tradeoff and compare our results with the reference paper by Gandor and Ksiazek. Our results demonstrate that classical feature-based methods can achieve competitive performance without deep learning.**

***Index Terms*—diabetic retinopathy, GLCM, genetic algorithm, feature selection, LightGBM, medical image analysis**

I. Introduction

Diabetic retinopathy is one of the most common complications of diabetes and is a leading cause of preventable blindness worldwide. According to recent estimates, around 470 million people suffer from diabetes globally, and that number is expected to keep rising [7]. The condition develops when high blood sugar damages blood vessels in the retina, and without treatment it can lead to permanent vision loss. Early detection is important, but in many parts of the world, especially in underserved regions, patients do not have easy access to ophthalmologists or diagnostic equipment. This creates a real need for automated systems that can help with screening.

Machine learning has been applied to a number of medical image analysis problems, including cancer detection, ECG analysis, and glaucoma screening. For diabetic retinopathy specifically, most recent work has relied on deep learning models like convolutional neural networks. While those methods can be very accurate, they also tend to need large datasets and a lot of compute power, which is not always available. An alternative approach is to use handcrafted features combined with classical machine learning. This is the direction we explored in this project, and it is motivated by the work of Gandor and Ksiazek [1], who showed that a pipeline using Gray-Level Co-occurrence Matrix (GLCM) features and a Genetic Algorithm (GA) can get competitive results on diabetic retinopathy prediction while being much lighter than deep learning.

Their paper, presented at GECCO 2025, used the APTOS 2019 Blindness Detection dataset [2] and proposed a framework that extracts Haralick texture features [3] from retinal fundus images using GLCM, then applies a genetic algorithm [5] to simultaneously tune the hyperparameters of a LightGBM classifier [4] and select the most relevant features. They tested several crossover strategies, and the best one, BLX-alpha-beta crossover, achieved 92.63% accuracy on the test set while reducing the feature set by 75%. The fitness function they used for the GA balances classification error against feature set size, and was adapted from Awadallah et al. [6]. Their results were competitive with several deep learning approaches on the same dataset.

Our project builds on this work. We follow the same general pipeline: image preprocessing, GLCM feature extraction, GA optimization, and LightGBM classification. But our implementation differs in a few ways. We use scikit-image instead of PyFeats for computing the GLCM, we extract 9 Haralick features at two window sizes (giving 18 total, compared to 14 in the paper), and we include a separate validation set to avoid any data leakage during optimization. We also added a baseline model, a default LightGBM trained on all 18 features with no optimization, so we could actually measure what the GA contributes. The goal is not just to reproduce the paper's pipeline but to evaluate the tradeoff between feature reduction and accuracy, and to understand which features the GA considers most important for this classification task.

II. Methods

A. Dataset

We used the APTOS 2019 Blindness Detection dataset [2], which is hosted on Kaggle and contains 3,662 retinal fundus images. Each image has a severity label from 0 to 4. For this project we converted these into binary labels: 0 for no diabetic retinopathy (originally label 0) and 1 for DR present (originally labels 1 through 4). After conversion the dataset was nearly balanced, with 1,805 healthy images and 1,857 DR images. This binary setup is the same as the one used in the reference paper [1].

B. Image Preprocessing

The raw APTOS images have large black background areas and vary quite a bit in quality. Our preprocessing pipeline handles this in four steps. First, the black background is removed by finding the largest contour in a thresholded grayscale version of the image and cropping to its bounding box. Second, CLAHE (Contrast Limited Adaptive Histogram Equalization) is applied to the L channel of the LAB color space, using a clip limit of 2.0 and tile grid size of 8×8. This improves local contrast without amplifying noise. Third, images are resized to 800×800 pixels. Fourth, they are converted to grayscale since GLCM works on single-channel images. Unlike the paper, We only preprocessed 1000 samples due to memory constraints.

Fig. 1. Raw (left) and preprocessed (right) retinal fundus image.

C. Data Splitting

We split the preprocessed images into training (70%, 700 images), validation (10%, 100 images), and test (20%, 200 images) sets. All splits are stratified to maintain class balance. The reference paper [1] used an 80/20 train/test split with no validation set, but we added one because the GA evaluates candidate solutions during optimization, and using the test set for that would be data leakage. We also set up 5-fold stratified cross-validation within the training set, which the GA uses internally to compute fitness scores. A fixed random seed of 42 was used throughout for reproducibility.

D. GLCM Feature Extraction

For each preprocessed image, we computed a Gray-Level Co-occurrence Matrix and extracted 9 Haralick texture features [3] from it: contrast, dissimilarity, homogeneity, energy, correlation, angular second moment (ASM), GLCM mean, GLCM variance, and entropy. These features capture different aspects of spatial texture in the image. This is relevant because diabetic retinopathy causes visible structural changes in the retina, like damaged blood vessels and microaneurysms, which affect image texture.

Following the reference paper [1], features were computed at two window sizes: 64×64 and 128×128 pixels. Each image was resized to the window size, quantized to 64 gray levels, and then the GLCM was computed at a distance of 1 pixel across four angles (0, 45, 90, and 135). Feature values were averaged across angles for rotation invariance. With 9 features at 2 window sizes, each image is represented by an 18-dimensional feature vector. We used scikit-image (`graycomatrix` and `graycoprops`) for the GLCM computation and Shannon entropy from `skimage.measure`.

E. Genetic Algorithm Optimization

The genetic algorithm [5] handles two tasks at once: it optimizes the hyperparameters of the LightGBM classifier [4] and selects the best subset of GLCM features. Each individual in the population is a real-valued chromosome with two sections. The first 18 genes correspond to the 18 features, where values in $[0.5, 1]$ mean a feature is selected and values in $[0, 0.5)$ mean it is rejected. The remaining 13 genes encode LightGBM hyperparameters (things like learning rate, max depth, number of estimators, etc.) scaled to their valid ranges.

We used BLX-alpha-beta crossover, which was the best performing strategy in the reference paper [1]. In this approach, offspring gene values are sampled uniformly from a range extended beyond the parents by randomly generated alpha and beta parameters. Tournament selection with $K = 3$ is used to pick parents, and each gene has a 20% chance of mutation (being replaced with a random value). The top individual from each generation is preserved through elitism.

The fitness function, adapted from Awadallah et al. [6], balances classification accuracy against model complexity:

$$\text{fitness} = \alpha \times (1 - \text{accuracy}) + \beta \times \frac{\text{selected\_features}}{\text{total\_features}} \quad (1)$$

We used $\alpha = 0.8$ and $\beta = 0.2$. The reference paper used 0.6/0.4, but in our early experiments those weights were too aggressive, and the GA cut the feature set down to just 3, which hurt accuracy noticeably. With 0.8/0.2, the GA still does meaningful feature selection but puts more weight on getting the accuracy right. We ran the GA with a population of 75 for 25 epochs, using 5-fold stratified cross-validation to evaluate each candidate.

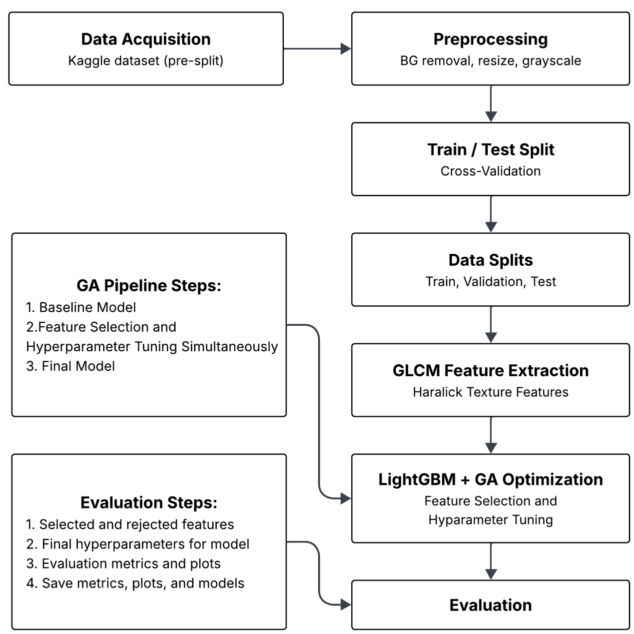

Fig. 2. End-to-end pipeline for diabetic retinopathy prediction.

F. Baseline Comparison

To properly evaluate what the GA adds, we also trained a baseline LightGBM classifier with default hyperparameters on all 18 GLCM features, with no optimization or feature selection. This gives us a reference point to see how much comes from the features themselves versus the GA's tuning and selection.

G. Evaluation Metrics

Both models were evaluated using accuracy, weighted precision, weighted recall, weighted F1-score, and ROC-AUC. We also generated confusion matrices. Since our dataset is nearly balanced, accuracy works fine as a primary metric, but the additional metrics give a fuller picture, especially in a medical context where missing a case of DR (a false negative) is particularly concerning.

H. Software and Tools

Everything was implemented in Python 3.12. The main libraries used were OpenCV for image preprocessing, scikit-image for GLCM computation, scikit-learn for splitting and evaluation metrics, LightGBM [4] for the classifier, NumPy and Pandas for data handling, and Matplotlib for plots. The code is available on GitHub at https://github.com/ayumahapat0/diabetic_retinopathy_prediction.

III. Results

A. Baseline Performance

The baseline LightGBM, trained with default parameters on all 18 features, achieved 91.00% test accuracy, 91.14% precision, 91.00% recall, and an F1 of 90.98%. The ROC-AUC on the test set was 0.959. On the validation set, accuracy was 87.19%. Looking at the confusion matrix (Fig. 3), the model correctly identified 317 out of 361 healthy images and 350 out of 372 DR images, with 44 false positives and 22 false negatives. For a model using only handcrafted texture features and no deep learning, this is a solid result.

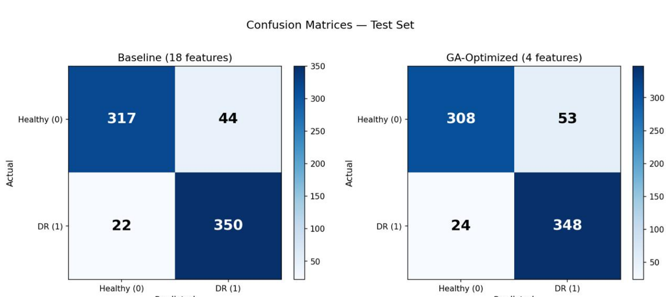

Fig. 3. Confusion matrices for the baseline and GA-optimized models on the test set.

B. GA-Optimized Performance

The GA selected 4 out of 18 features: correlation at 64×64, entropy at 64×64, contrast at 128×128, and ASM at 128×128. That is a 77.8% reduction in the feature set. With these 4 features and the tuned hyperparameters, the model achieved 89.50% test accuracy, 89.73% precision, 89.50% recall, 89.47% F1, and ROC-AUC of 0.948. On validation, accuracy was 85.56%.

The best hyperparameters the GA found were: 50 estimators, max depth of 11, learning rate of 0.217, 33 leaves, minimum child samples of 43, and regularization lambda of 9.74. The GA converged to its best fitness score of 0.1253 by epoch 7 out of 25, and the fitness stayed flat after that, as shown in Fig. 4. This means the algorithm found a stable solution fairly early.

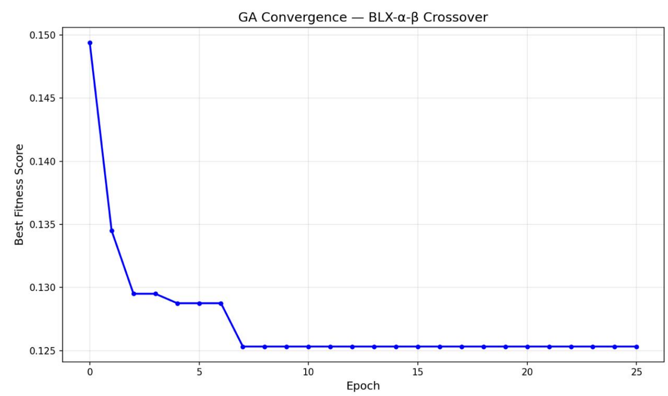

Fig. 4. GA convergence over 25 epochs. The fitness stabilized by epoch 7.

C. Baseline vs. GA Comparison

Table I summarizes the comparison between the two models. The baseline outperforms the GA-optimized model by about 1.5% across all metrics. However, the GA model uses only 4 features compared to the baseline's 18. The ROC curves in Fig. 5 show that both models perform well above random, with AUC values of 0.959 and 0.948 respectively. Fig. 6 shows the metric comparison as a bar chart.

TABLE I
BASELINE AND GA-OPTIMIZED MODELS COMPARISON ON THE TEST SET

Model	*Acc.*	*Prec.*	*Rec.*	*F1*	*AUC*
Baseline (18 feat.)	0.9100	0.9114	0.9100	0.9098	0.9590
GA-Opt. (4 feat.)	0.8950	0.8973	0.8950	0.8947	0.9480

D. Feature Selection Analysis

The four features that the GA kept are worth looking at. Correlation at 64×64 captures linear dependencies between neighboring pixel intensities, which could reflect the regularity of retinal structures. Entropy at 64×64 measures randomness in the texture, which might be higher in diseased retinas because of irregular vascular patterns. Contrast at 128×128

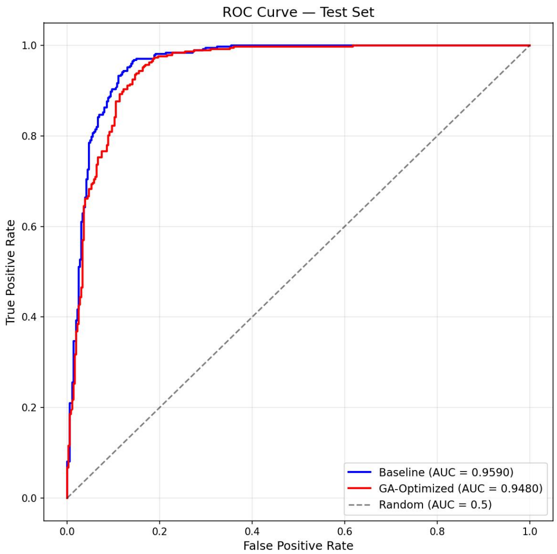

Fig. 5. ROC curves for both models on the test set.

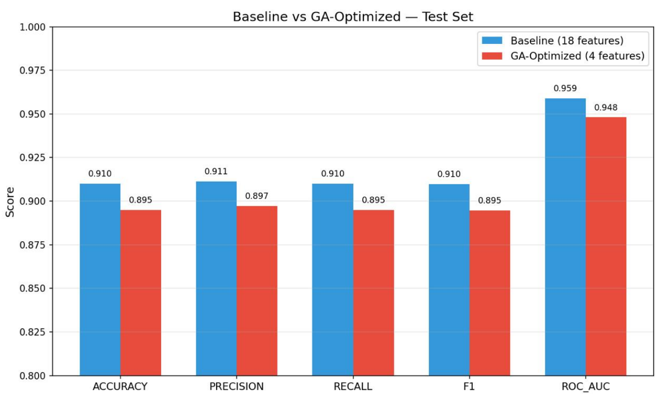

Fig. 6. Side-by-side comparison of evaluation metrics.

picks up intensity differences at a larger scale, and ASM at 128×128 measures texture uniformity. The fact that the GA chose features from both window sizes suggests both local and broader spatial patterns matter for this task. Fig. 7 shows which features were kept and which were dropped.

Some features that might seem useful at first glance, like homogeneity and energy, were rejected. This does not mean they are uninformative by themselves, but the GA likely found them redundant given the other selected features, or the small improvement they offered was not worth the added complexity penalty in the fitness function.

E. Comparison with Reference Paper

The reference paper [1] reported 92.63% accuracy with BLX-alpha-beta crossover and a 75% feature reduction. Our baseline gets 91.00% and the GA-optimized model gets 89.50% with 77.8% reduction. A few things explain the gap. First, the paper used 14 Haralick features (computed with PyFeats) versus our 9 (from scikit-image). Second, they ran the

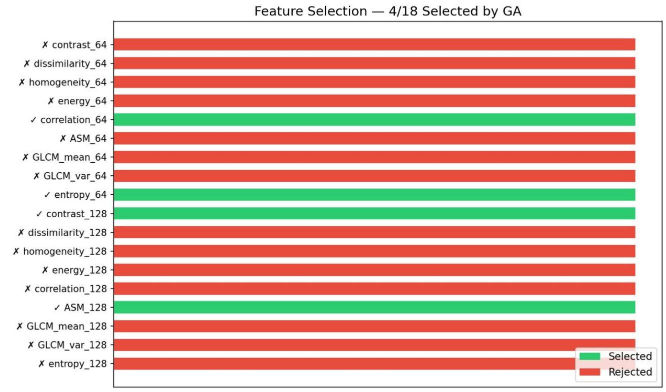

Fig. 7. Feature selection results. 4 out of 18 GLCM features were retained.

GA for 100 epochs with a population of 100, while we used 25 epochs and 75 individuals. Third, they used the original 0.6/0.4 fitness weights. They also repeated each experiment 10 times and reported the best result across all runs. Our numbers come from a single run with a fixed seed.

IV. Conclusions

This project shows that a classical, feature-based approach to diabetic retinopathy detection can produce solid results without deep learning. Our baseline LightGBM, using all 18 GLCM texture features, hit 91.00% test accuracy and a ROC-AUC of 0.959. That is close to the 92.63% from the reference paper [1] and competitive with several deep learning methods on the APTOS dataset [7].

The GA-optimized model demonstrates the accuracy versus complexity tradeoff. Cutting the feature set from 18 to 4 gives a much simpler model at the cost of about 1.5% accuracy. Whether that tradeoff makes sense depends on context. In a clinical screening environment where compute is limited and interpretability matters, a 4-feature model with 89.5% accuracy and 0.948 AUC might be more practical than a full model or a deep learning solution.

There are some clear limitations. GLCM features only capture texture and miss other visual markers of DR like microaneurysms, hemorrhages, or hard exudates. The binary classification (DR vs. no DR) is a simplification of the real clinical problem, where severity grading matters for treatment. And our GA ran with fewer resources than the reference paper, so it may not have found the best possible solution. More epochs, a larger population, and more samples could potentially close the gap.

For future work, a few directions seem promising. Using a larger set of Haralick features (closer to the 14 used in the paper) would give the GA more material to work with. Testing additional crossover strategies (the paper compared five different ones) could help too. A hybrid approach combining GLCM features with deep learning features might get the best of both worlds. And extending the pipeline to multi-class severity prediction instead of binary would make it more useful in a real clinical setting.

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